# Chapter 5 Fundamentals of Machine Learning and Deep Learning

High-level definitions of fundamental concepts

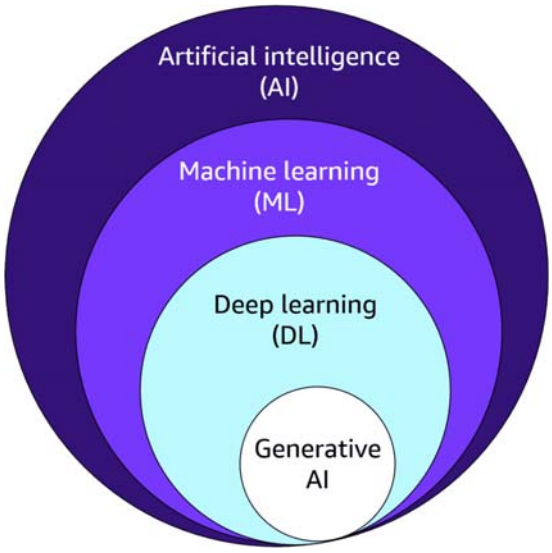
Timeline of the development of machine learning & deep learning

Key factors behind machine learning

Asst. Prof.Rapeeporn Chamchong, PhD
Department of Computer Science, Faculty of Informatics, MSU



## AI vs ML vs DL




AI: A technique enables machines to mimic human behavior.

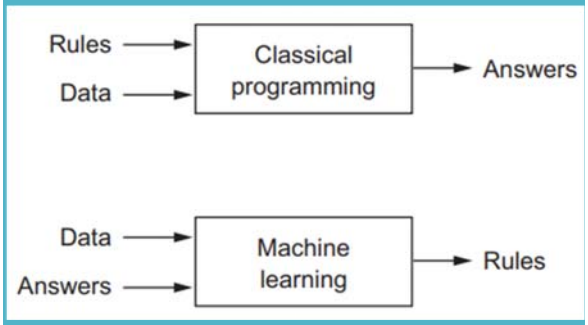
ML: A subset of AI utilizes statistical methods to help machines learn from datasets and make predictions.

DL: A subset of ML that makes the computation of a multi-layer neural network feasible.

GenAI: A subset of DL that learns patterns from existing training data and generates new output.

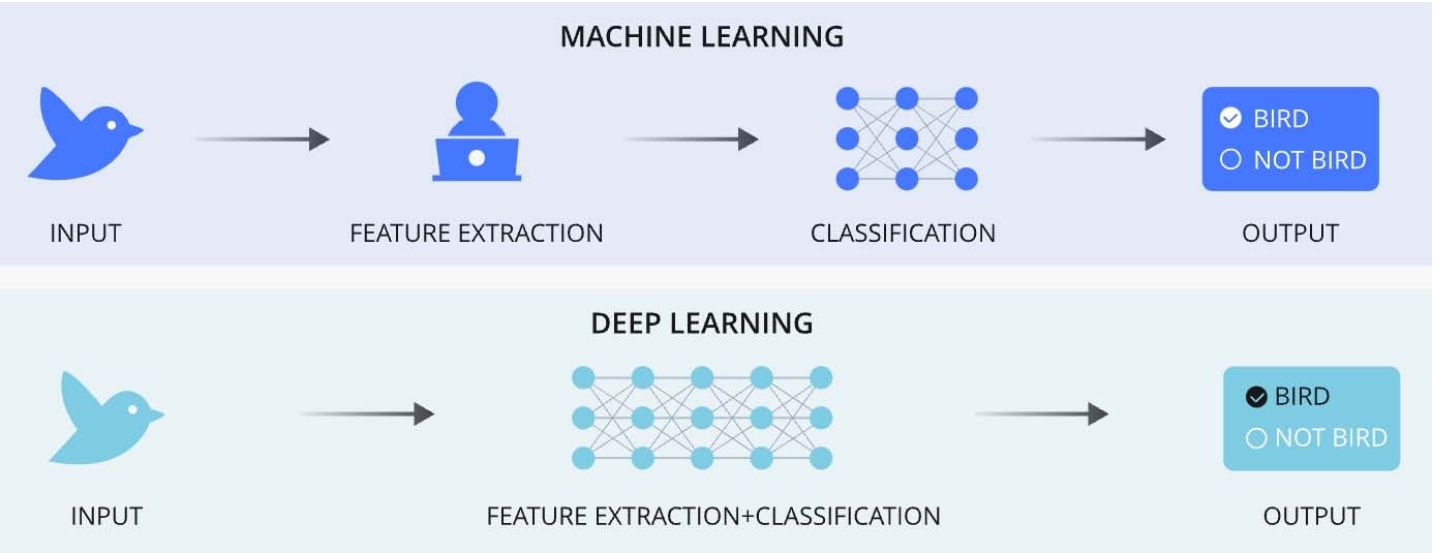
- In classical programming (AI), humans input rules (a program) and data to be processed according to these rules, and the outcome answers.
- A ML system is trained with many examples rather than explicitly programmed.

## AI & ML





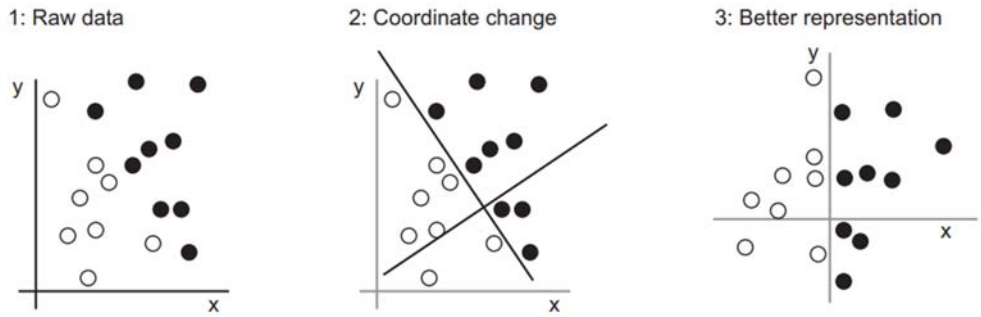

## ML vs DL




Source: https://www.fs.com/blog/artificial-intelligence-vs-machine-learning-vs-deep-learning-vs-generative-ai-key-differences-explained-24134.html

# ML & DL

- ML is all about finding appropriate representations for input data (transformations of data-eg. classification task)
  - Input data points
  - Examples of the expected output
  - A way to measure whether the algorithm is doing a good job

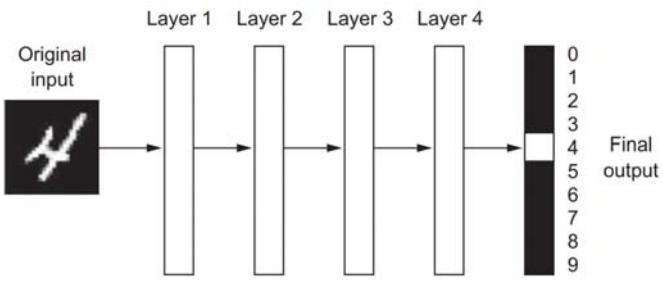


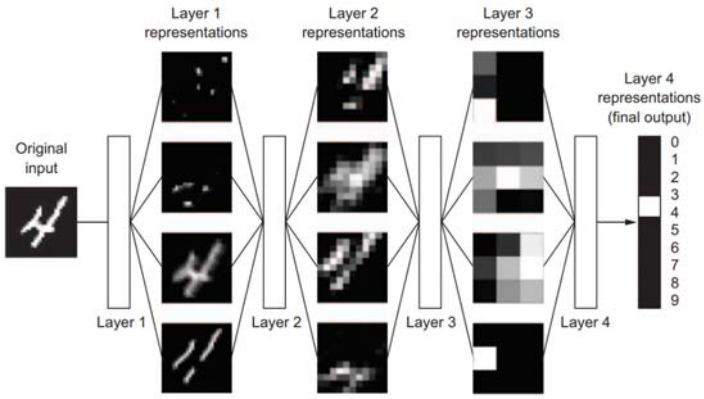

Source: François Chollet, Deep Learning with Python, Manning Publications Co., 2018



# ML & DL

- Deep learning is a specific subfield of machine learning:
  - Learning representations from data
    - *learning successive layers of increasingly meaningful representations*
- DL is a mathematical framework for learning representations from data
  - What do the representations learned by a deep-learning algorithm look like?

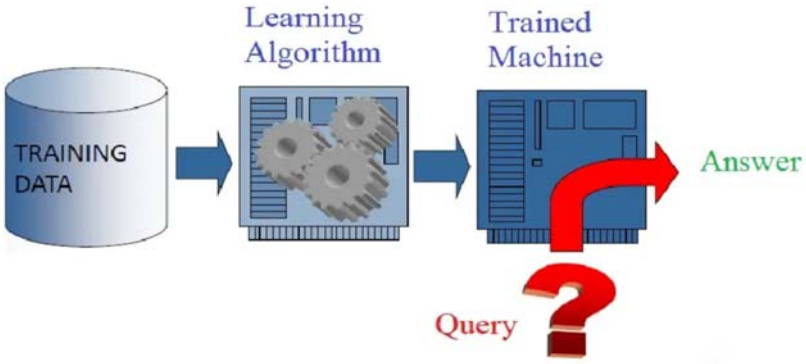



Source: François Chollet, Deep Learning with Python, Manning Publications Co., 2018

# Machine Learning (ML)

- The goal of ML is to learn from data
  - Technically, combines statistics and computational tools (optimization)
  - Using algorithm that iteratively learn from data, machine learning allows computer to find hidden insights without being explicitly programmed where to look.

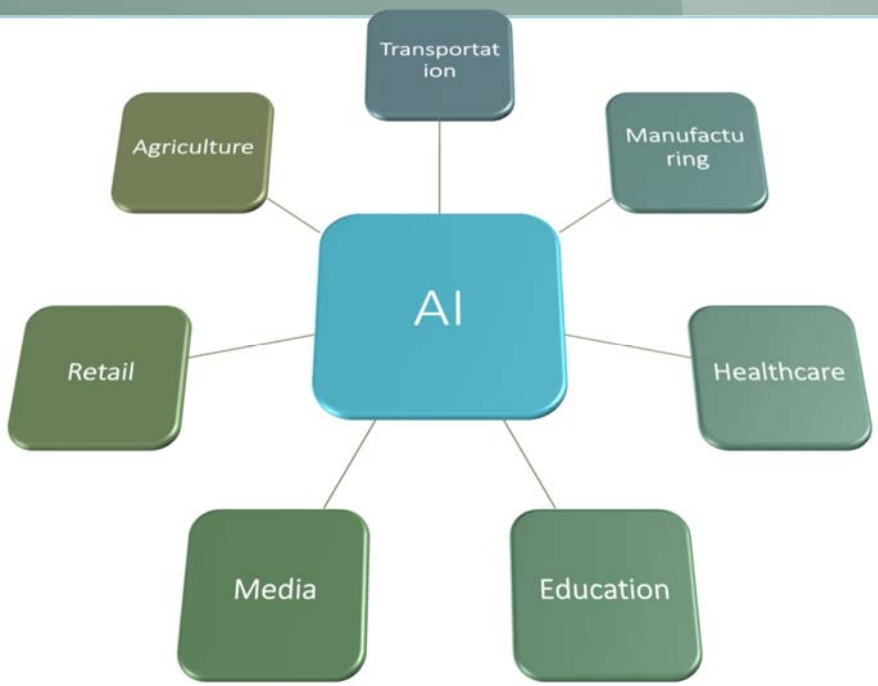





# The main advantage of ML

- Learning and writing an algorithm
  - It's easy for the human brain, but it is tough for a machine.
  - It takes some time and a good amount of training data for the machine to accurately classify objects
- Implementation and automation
  - This is easy for a machine.
  - Once learnt a machine can process one million images without any fatigue, where as human brain can't.
- That's why ML with big data is a deadly combination

# Impact of AI






# Agriculture

SINGAPORE VERTICAL FARMING START-UP EXPANDS TO HONG KONG

**AGRIONE**

IoT-enabled AI-driven Smart Agriculture

# Manufacturing

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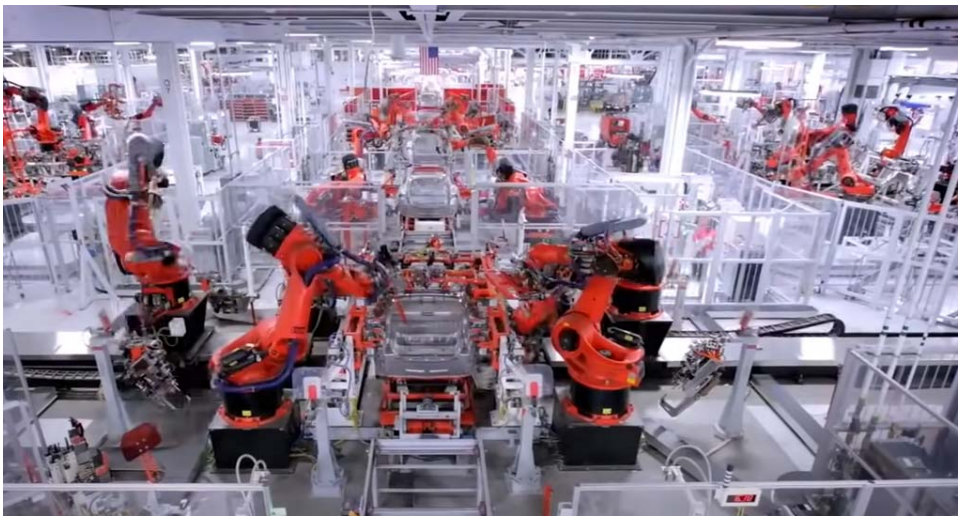
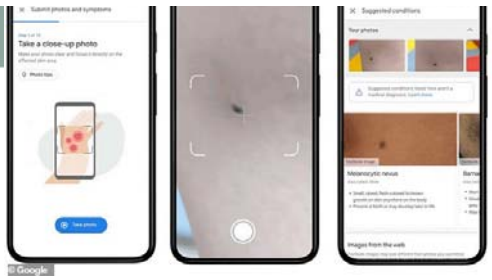


https://www.youtube.com/watch?v=uobJbNyGVjo&ab_channel=CasgainsCapital



# Healthcare

Google Skin Condition Prediction



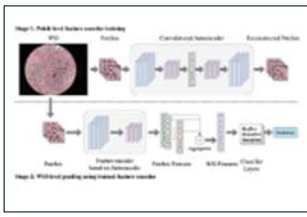
https://australiannewsreview.com/google-unveils-ai-tool-that-can-identify-288-skin-conditions/

**MeMoSA® – mobile mouth screening anywhere**

cancer research malaysia

*Mouth cancer screening – anywhere and anytime. Technology has supported many advances in the field of science and medicine and it's about to do even more for mouth cancer. Find out how you can be part of this, and work with us towards a future free of the fear of cancer.*

# Healthcare – 2021's papers



**Grading Invasive Ductal Carcinoma from Whole-Slide Histological Images using Deep Learning-Based Feature Encoding Techniques**

Zhiguo Huang, Chatklaw Jareanpon, Phatthanaphong Chomphuwiset, Rapeeporn Chamchong 125-138

PDF doi: 10.14456/mijet.2026.12

**Deep learning** and medical image processing for coronavirus (COVID-19) pandemic: A survey.

Bhattacharya S, et al. Sustain Cities Soc. 2021. PMID: 33169099

Free PMC article.

Numerous sources of medical images (e.g., X-ray, CT, and MRI) make **deep learning** a great technique to combat the COVID-19 outb ...

Anti-senescent drug screening by **deep learning**-based morphology senescence scoring.

Kusumoto D, et al. Nat Commun. 2021. PMID: 33431893

Free PMC article.

Advances in **deep learning** technology have enabled complex task solutions. The accuracy of image classification tasks has impro ...

Cite Share

**Deep learning** model for the prediction of microsatellite instability in colorectal cancer: a diagnostic study.

Yamashita R, Long J, Longacre T, Peng L, Berry G, Martin B, Higgins J, Rubin DL, Shen J.

Lancet Oncol. 2021 Jan;22(1):132-141. doi: 10.1016/S1470-2045(20)30535-0.



# Retails

Amazon.com claims that 35% of its revenue is generated by its recommendation engines, one of the most popular and upcoming concepts in AI (source). It is also referred to by marketers as **Real Time Product**

https://usmsystems.com/top-use-cases-of-artificial-intelligence-in-retail-industry/

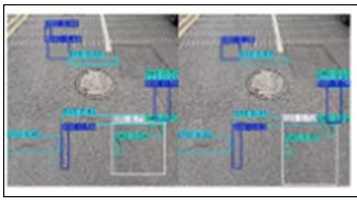
https://medium.com/@humansforai/retail-d4ca38b4de3f

# Transportation

Tokyo – Autonomous Taxi

Waymo company - self-driving minivans and trucks

# Transportation - Traffic

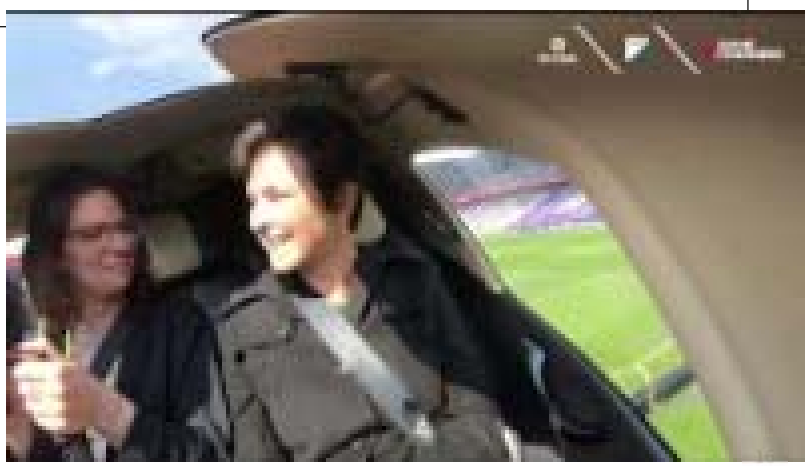


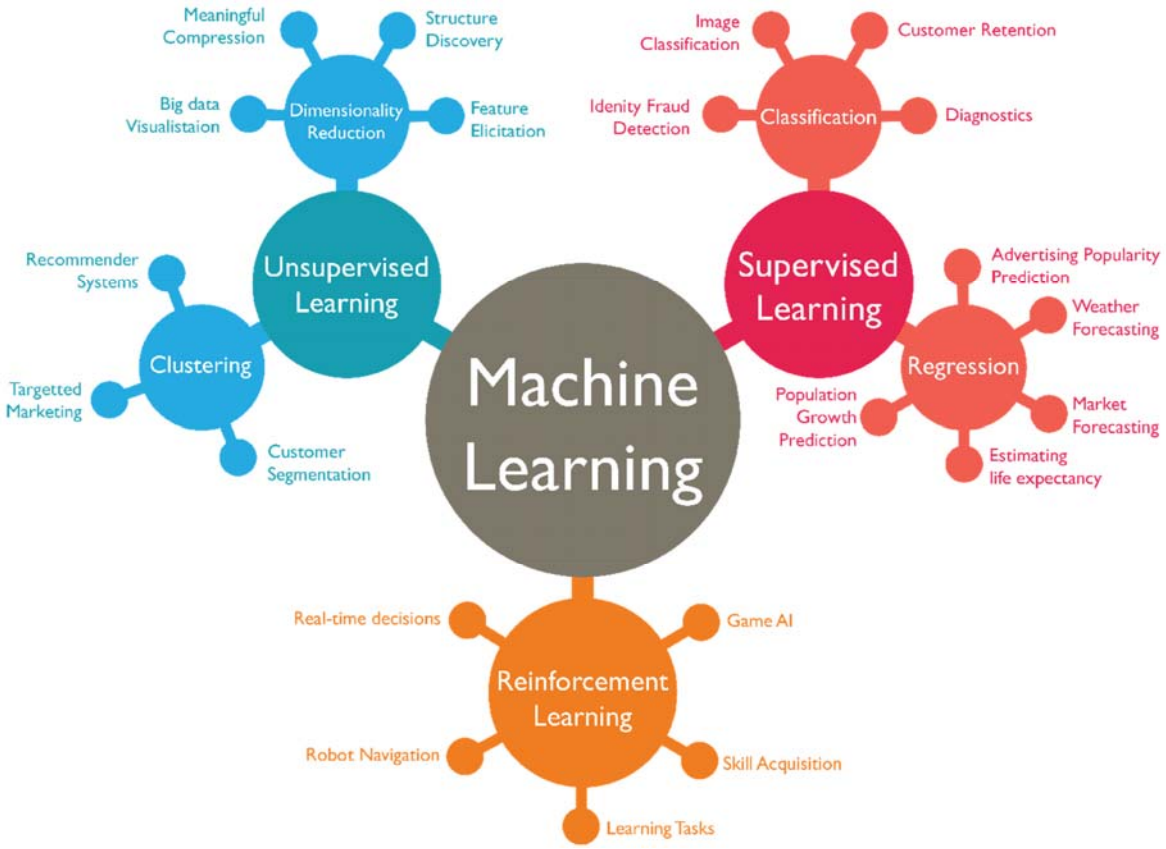
**Applied Ensemble Technique for Road Damage Detection based on YOLOv8**

Zhipeng Tang, Apirak Jirayusakun, Rapeeporn Chamchong 135-147

PDF doi: 10.14456/mijet.2026.13

Drone Taxi








# The Types of Machine Learning Algorithms

**ML**

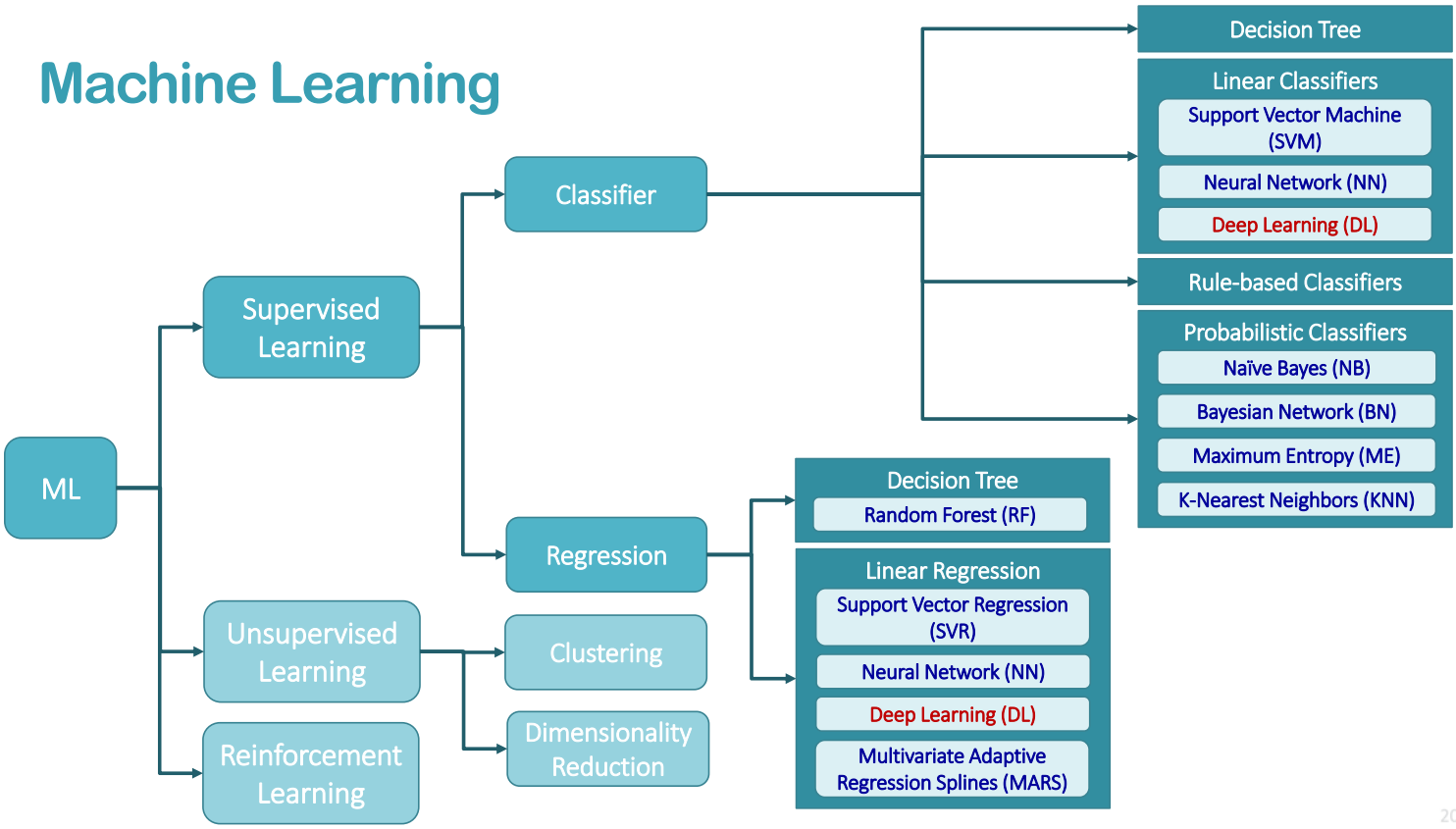
| Supervised Learning | Unsupervised Learning | Reinforcement Learning |
|---|---|---|
| **Task driven**<br>- **Prediction**<br>- **Classification (discrete labels)**<br>- **Regression (real values)** | **Data driven**<br>- **Clustering**<br>- **Probability distribution**<br>- **Finding association**<br>- **Dimension reduction** | **Algorithm learns to react to environment**<br>**- Decision making (robot, chess machine)** |
| • develop a prediction model based on input & output (label).<br>• Goal is to learn mapping input & output data | • Group/interpret data based only on input data<br>• No labeled example outputs | • Dynamic environment<br>• Trial and error which actions yield the greatest rewards |

# Machine Learning

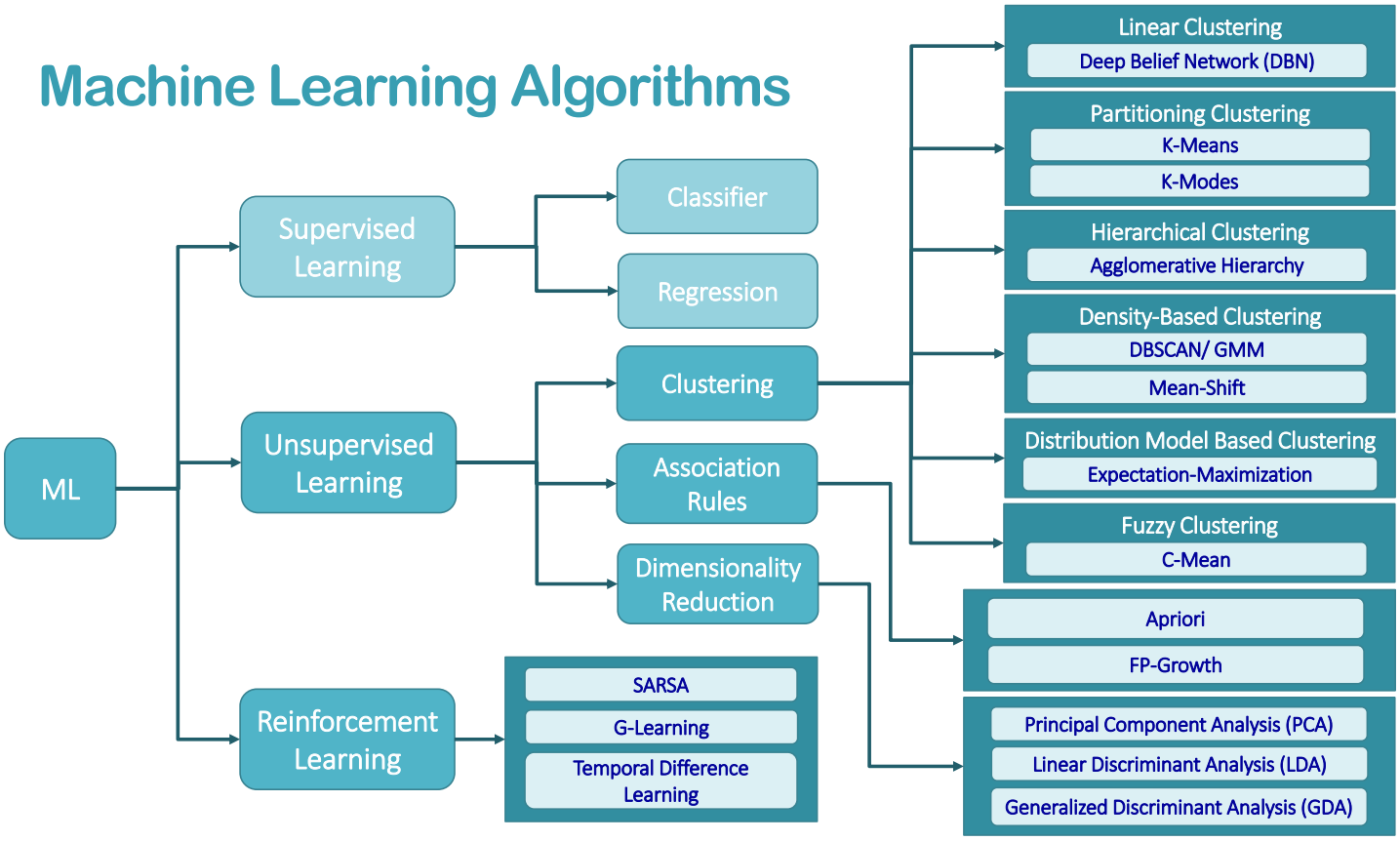
- Learning paradigms
  - Supervised Learning
    - *The computer is presented with example inputs and their desired outputs, given by a "teacher", and the goal is to learn a general rule that maps inputs to outputs.*
  - Unsupervised Learning
    - *No labels are given to the learning algorithm, leaving it on its own to find structure in its input. Unsupervised learning can be a goal in itself (discovering hidden patterns in data) or a means towards an end (feature learning).*
  - Reinforcement Learning
    - *A computer program interacts with a dynamic environment in which it must perform a certain goal (such as driving a vehicle or playing a game against an opponent). As it navigates its problem space, the program provides feedback that's analogous to rewards, which it tries to maximize.*



# Machine Learning

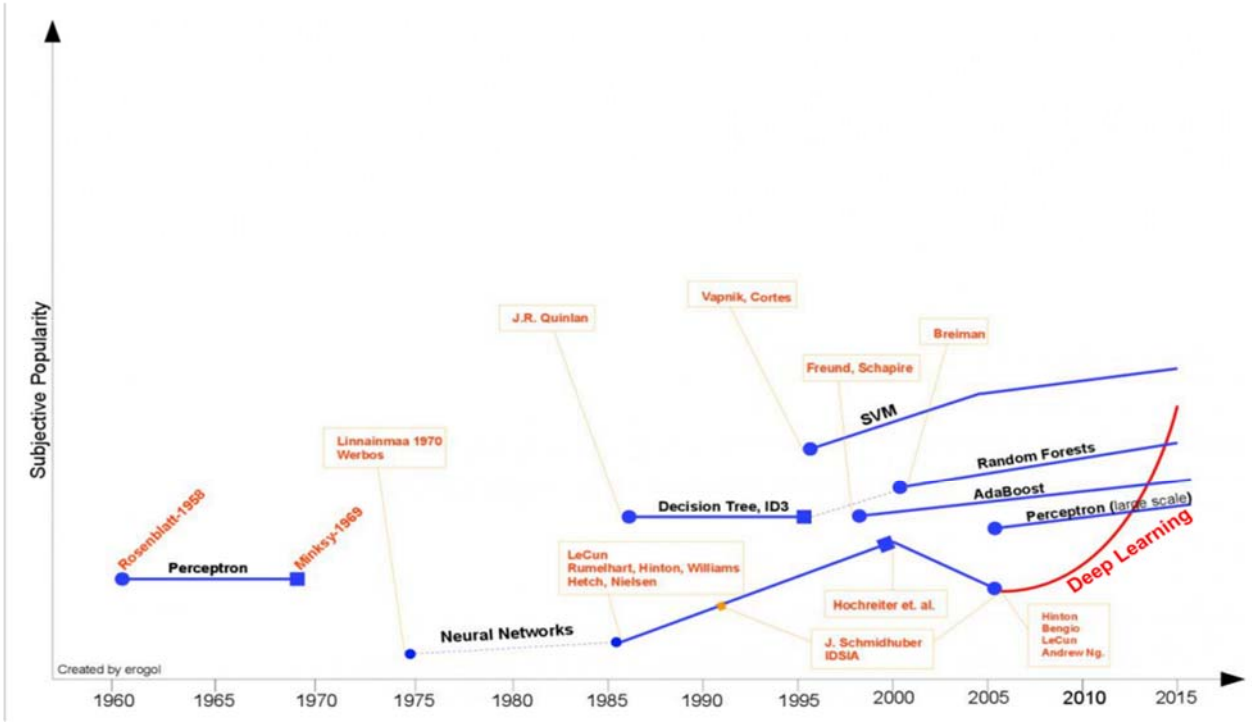


# Machine Learning Algorithms






# ML & DL Evolution



# Example (supervised learning) tasks

- Use in a marketing company is trying to find out which customers are likely to churn.
- Use it to predict the likelihood of occurrence of perils like earthquakes, tornadoes, weather, etc.
- Determine the Total Insurance Value.

→ Model → Cat or Dog

Image recognition / classification

→ Model → "Hello"

Speech recognition

"How are you?" → Model → "คุณเป็นอย่างไร?"

Machine translation

"How are you?" → Model → "I am fine thank you"

Conversational agent / chatbot



# Example (unsupervised learning) tasks

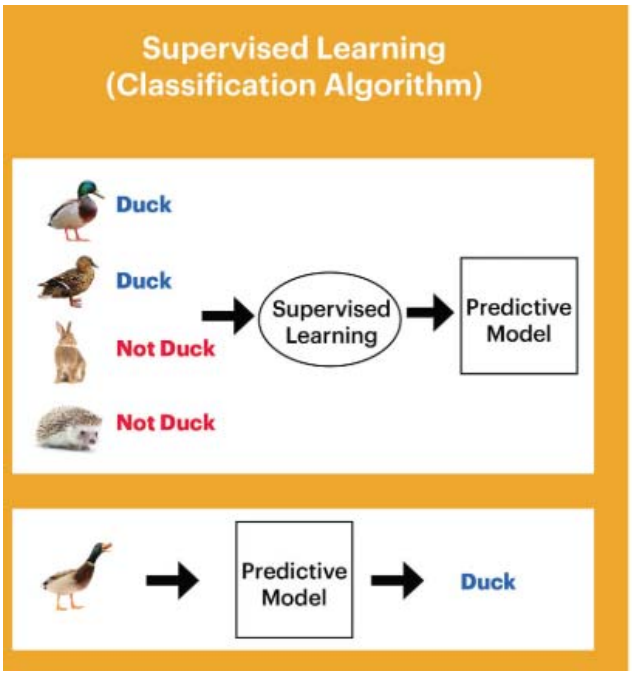
- Association/ Recommendation
  - when a retailer wishes to find out what is the combination of products, customers tend to buy more frequently such as in Amazon, Shopee, Lazada
  - in the pharmaceutical industry, unsupervised learning may be used to predict which diseases are likely to occur along with diabetes
  - Amazon, recommends the product
- Clustering/ Dimension Reduction
  - Grouping the documents or grouping customers
  - Image Segmentation

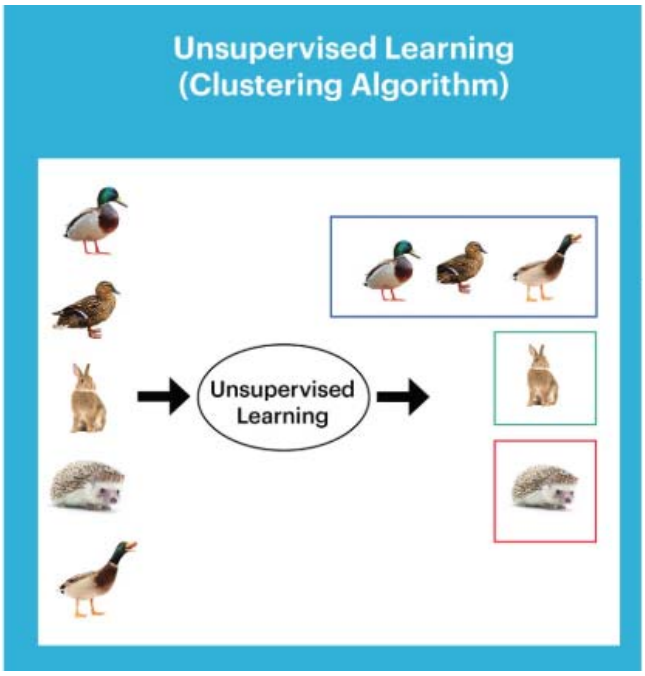
# Reinforcement Learning

- Algorithm discovers through trial and error which actions yield the greatest rewards.
- Three primary components:
  - the agent (the learner or decision maker),
  - the environment (everything the agent interacts with)
  - actions (what the agent can do).
- Objective: the agent chooses actions that maximize the expected reward over a given amount of time.



# Example of Supervised VS Unsupervised ML

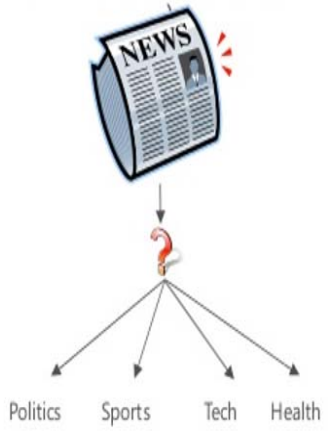


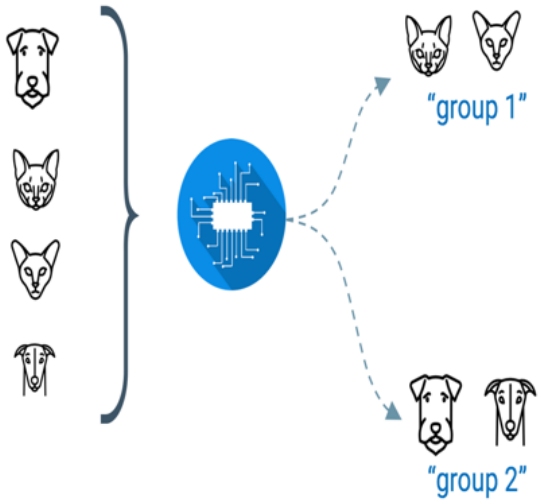
# Example of Supervised VS Unsupervised ML

Supervised

Classify a news article as (politics, sports, technology, health,




Unsupervised






# Example of Supervised VS Unsupervised ML

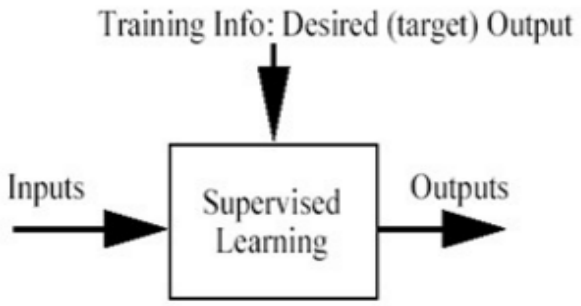
Supervised

| sepal_length | sepal_width | petal_length | petal_width | species |
|---|---|---|---|---|
| 4.6 | 3.2 | 1.4 | 0.2 | Iris-setosa |
| 5.3 | 3.7 | 1.5 | 0.2 | Iris-setosa |
| 5 | 3.3 | 1.4 | 0.2 | Iris-setosa |
| 7 | 3.2 | 4.7 | 1.4 | Iris-versicolor |
| 6.4 | 3.2 | 4.5 | 1.5 | Iris-versicolor |
| 6.9 | 3.1 | 4.9 | 1.5 | Iris-versicolor |
| 5.5 | 2.3 | 4 | 1.3 | Iris-versicolor |

Unsupervised

| Customer Id | Age | Edu | Years Employed | Income | Card Debt | Other Debt | Address | DebtIncomeRatio |
|---|---|---|---|---|---|---|---|---|
| 1 | 41 | 2 | 6 | 19 | 0.124 | 1.073 | NBA001 | 6.3 |
| 2 | 47 | 1 | 26 | 100 | 4.582 | 8.218 | NBA021 | 12.8 |
| 3 | 33 | 2 | 10 | 57 | 6.111 | 5.802 | NBA013 | 20.9 |
| 4 | 29 | 2 | 4 | 19 | 0.681 | 0.516 | NBA009 | 6.3 |
| 5 | 47 | 1 | 31 | 253 | 9.308 | 8.908 | NBA008 | 7.2 |
| 6 | 40 | 1 | 23 | 81 | 0.998 | 7.831 | NBA016 | 10.9 |
| 7 | 38 | 2 | 4 | 56 | 0.442 | 0.454 | NBA013 | 1.6 |
| 8 | 42 | 3 | 0 | 64 | 0.279 | 3.945 | NBA009 | 6.6 |
| 9 | 26 | 1 | 5 | 18 | 0.575 | 2.215 | NBA006 | 15.5 |
| 10 | 47 | 3 | 23 | 115 | 0.653 | 3.947 | NBA011 | 4 |
| 11 | 44 | 3 | 8 | 88 | 0.285 | 5.083 | NBA010 | 6.1 |
| 12 | 34 | 2 | 9 | 40 | 0.374 | 0.266 | NBA003 | 1.6 |

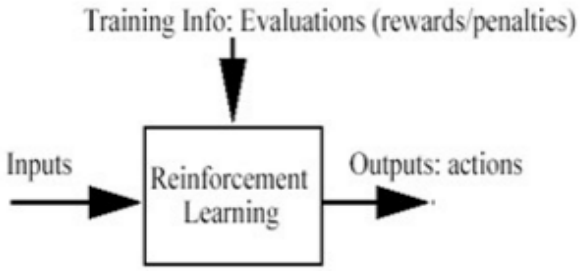
## Supervised Learning




Error = (target output - actual output)

Input is an istance, output is a classification of the istance

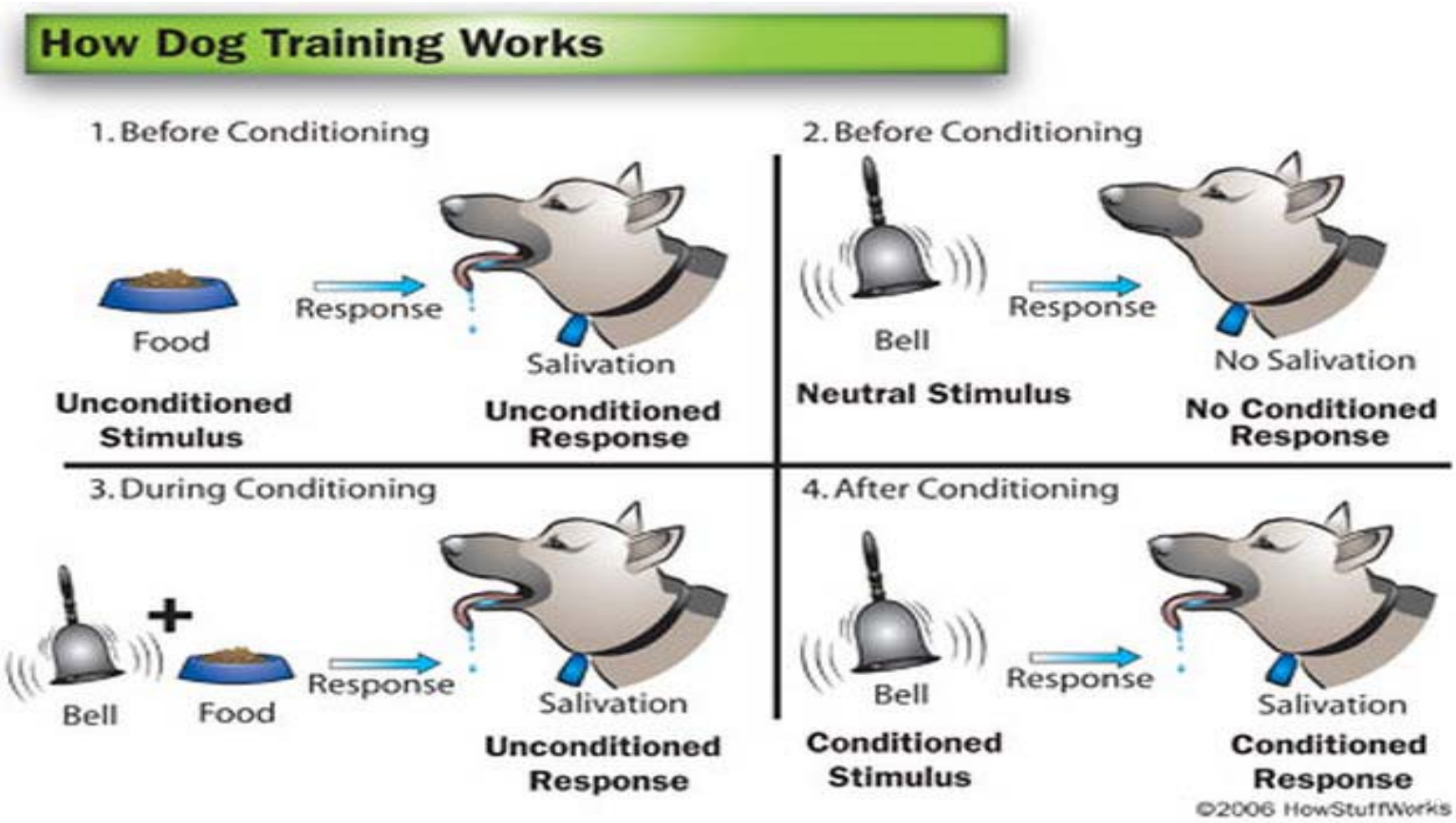
## Reinforcement Learning



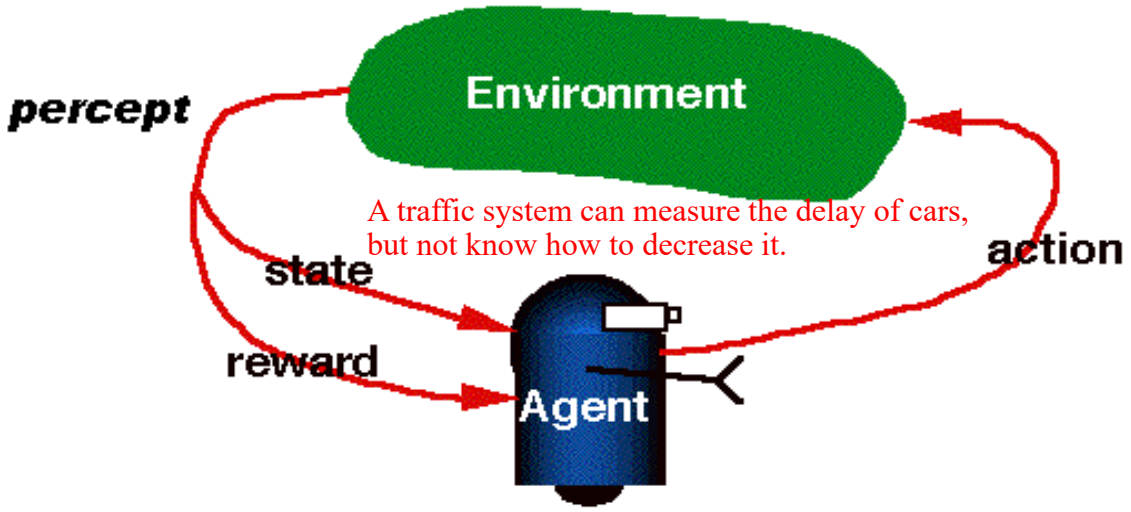

*Objective: Get as much reward as possible*

Input is some "goal" output
Is a sequence of actions to meet the goal





# RL is learning from interaction






# Example (reinforcement learning) tasks

Online Poker

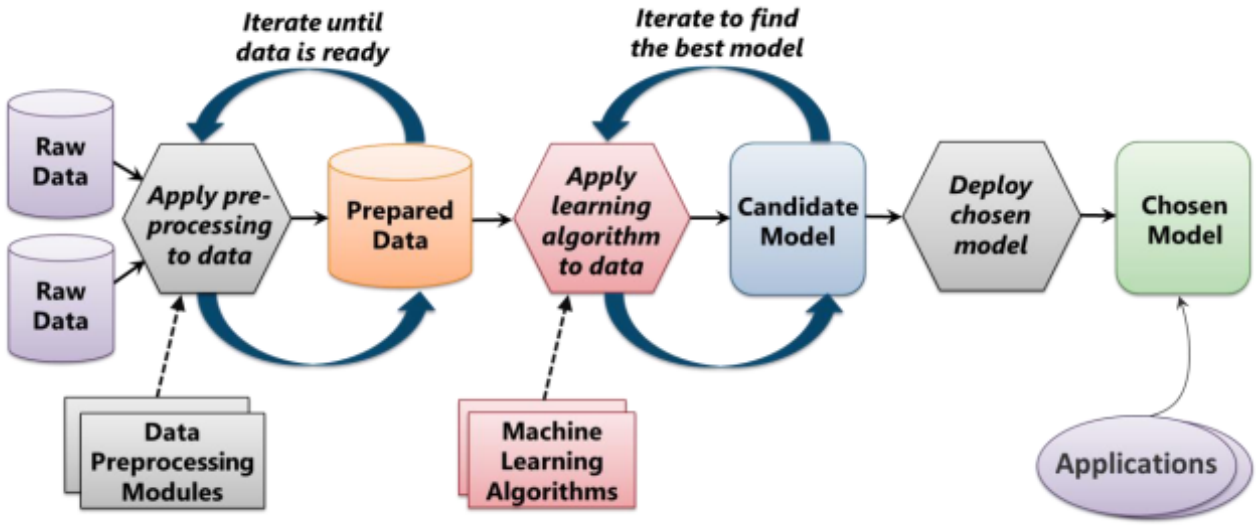
Source:
*RoboCup web site*

# Comparing Types of ML

| Supervised learning | Unsupervised learning | Reinforcement learning |
|---|---|---|
| Input/Output pairs | Input only | Input & Critic |
| Learning phase v.s. acting phase | Learning phase v.s. acting phase | Learning and acting simultaneously – online learning |
| | | Explicitly explore the world and learn by trial and error |



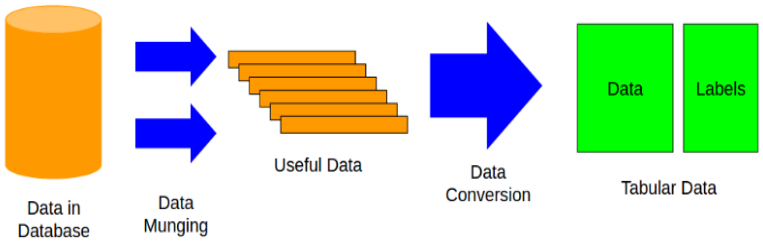
# Typical Machine Learning Process




Source: INTRODUCING AZURE MACHINE LEARNING, pg. 5

# Typical Machine Learning Process (cont.)

- Data collection:
  - Be it the raw data from excel, access, text files etc.
  - This step (gathering past data) forms the foundation of the future learning. The better the variety, density and volume of relevant data, better the learning prospects for the machine becomes.
- Data preprocessing:
  - Any analytical process thrives on the quality of the data used.
    - *Selecting data*
    - *Cleaning data such as missing values or outliers*
    - *Reducing data*



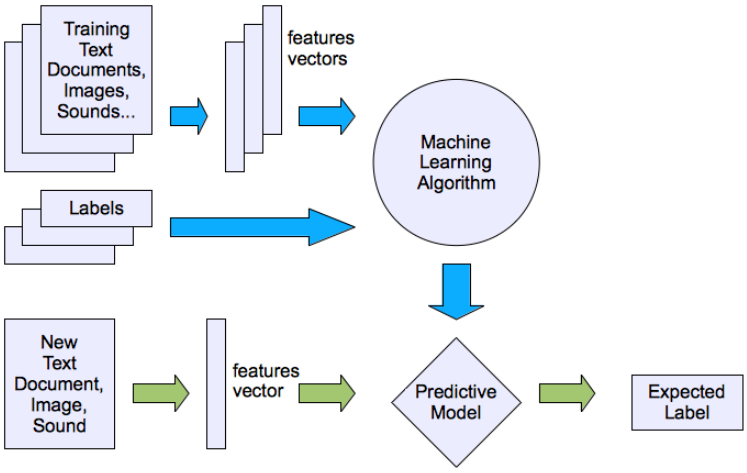



# Typical Machine Learning Process (cont.)

- Training a model:
  - Choosing the appropriate algorithm and representation of data in the form of the model.
  - The cleaned data is split into two parts – train and test (proportion depending on the prerequisites);
    - *the first part (training data) is used for developing the model.*
    - *The second part (test data), is used as a reference.*
- Evaluating the model:
  - To test the accuracy, the second part of the data (holdout / test data) is used.
  - This step determines the precision in the choice of the algorithm based on the outcome.
  - A better test to check accuracy of model is to see its performance on data which was not used at all during model build.
- Improving the performance:
  - This step might involve choosing a different model altogether or introducing more variables to augment the efficiency.
  - That's why significant amount of time needs to be spent in data collection and preparation.

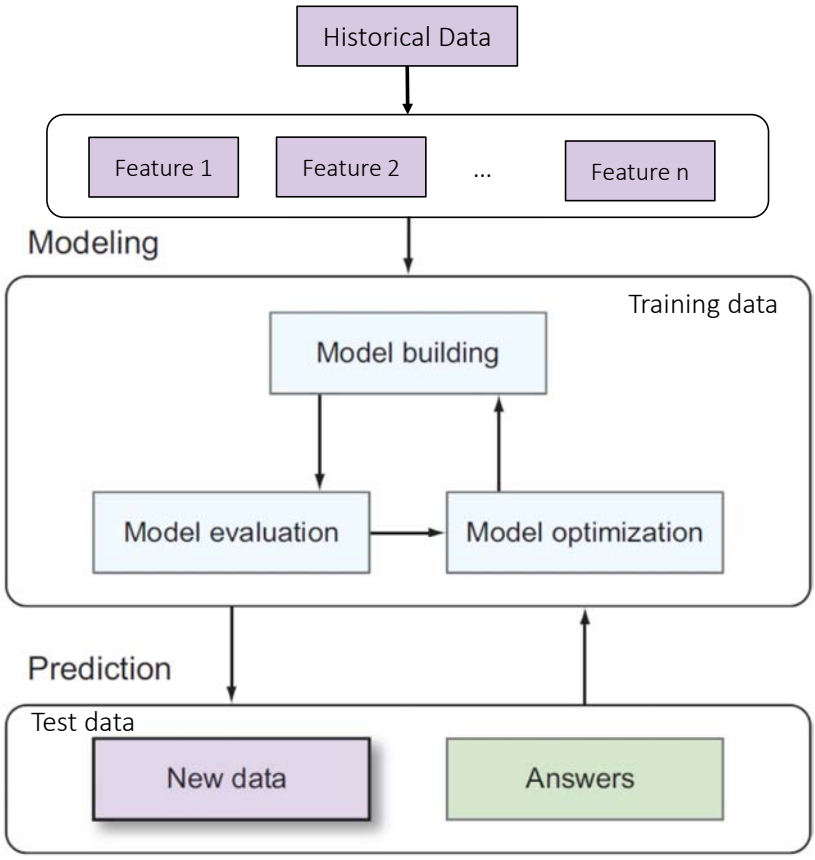
# Supervised Learning

- Trained using labeled examples
- Desired output is known
- Uses patterns to predict the values of the label on additional unlabeled data
- Methods include classification, regression, etc.
- Some examples of algorithms used are Nearest neighbor, Naïve Bayes, Decision Trees, Regression Neural *Networks, SVM, etc.*




Source: : Yi-Fan Chang, An overview of machine learning (slideshare.net)

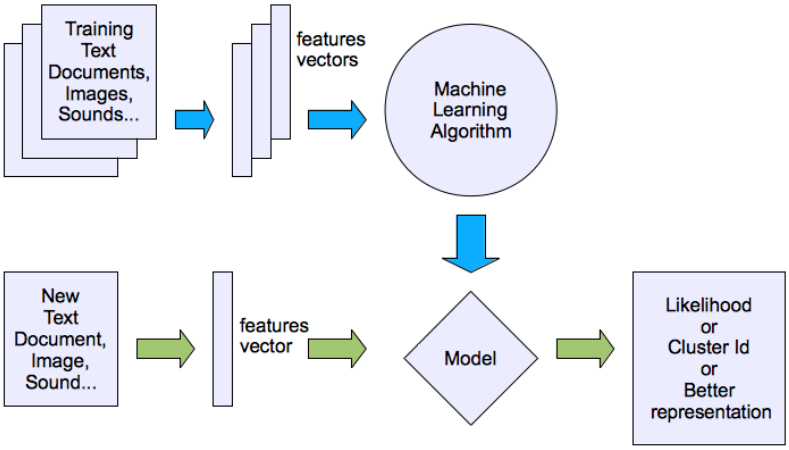





# Supervised Learning Workflow

# Unsupervised Learning

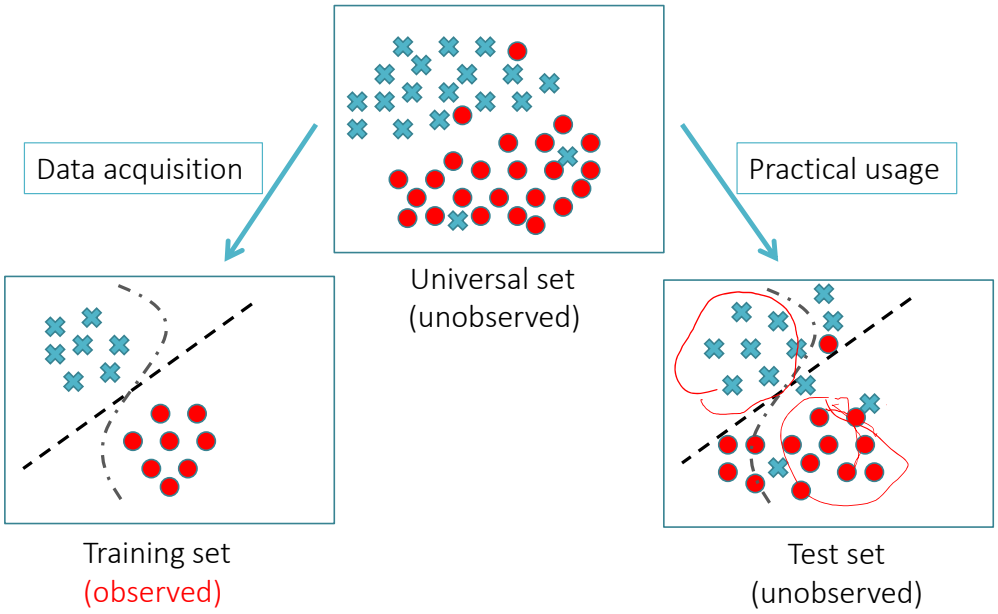
- Used against data that has no historical labels
- The system is not told the "right answer"
- Goal is to explore the data and find some structure within the data
- Methods
  - Clustering: K-mean, C-mean
  - Feature selection (Dimension reduction): PCA, Information Gain
  - Density estimation-Gaussian Mixture Model (GMM), Graphical models
  - Finding association (in features)



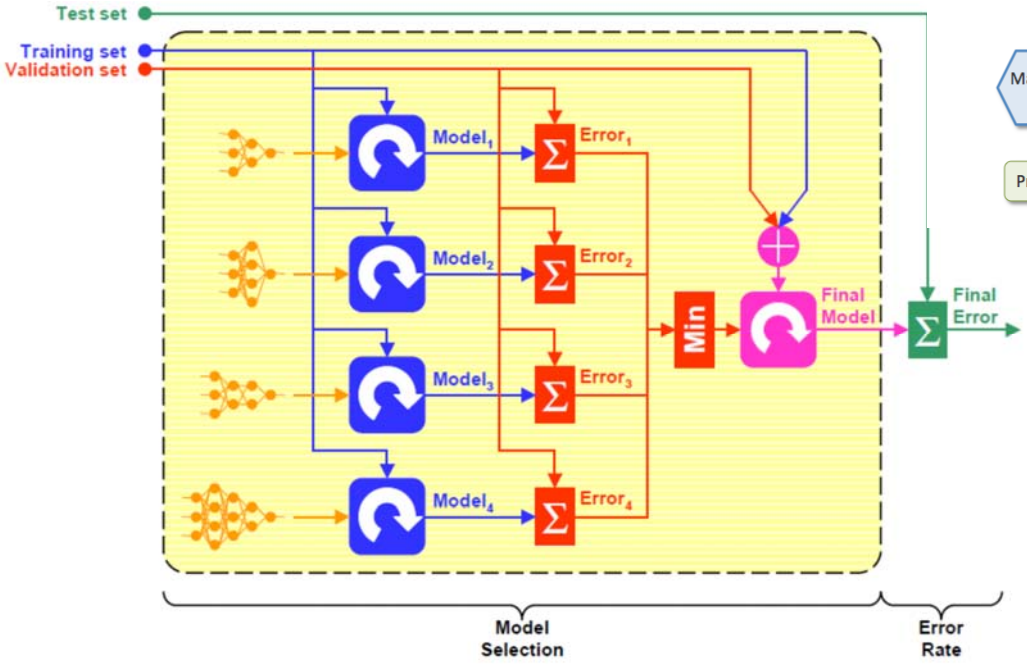

Source: Yi-Fan Chang, An overview of machine learning (slideshare.net)

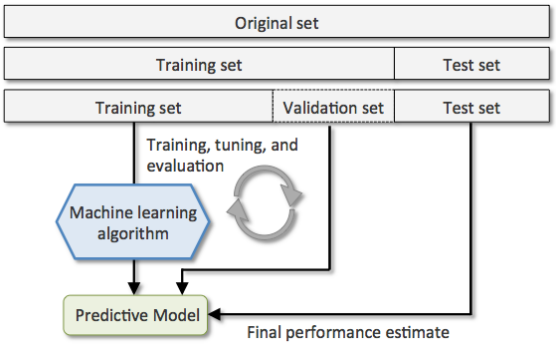


# Training and Test Set



# Three-way Data Splits



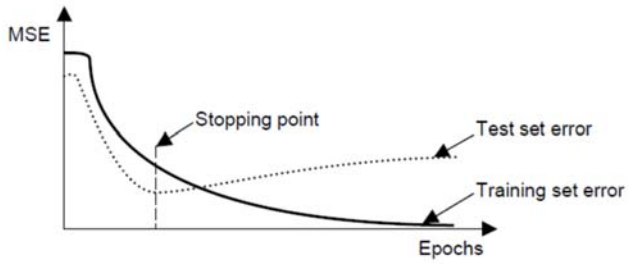





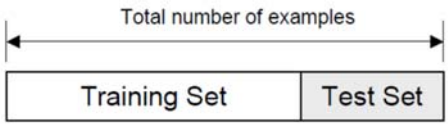

# How to Separate Train/Test Dataset

- Hold-out method
- Cross validation
  - Random Subsampling
  - K-Fold Cross-Validation
  - Leave-one-out Cross-Validation
- Bootstrap

# Hold-out Method



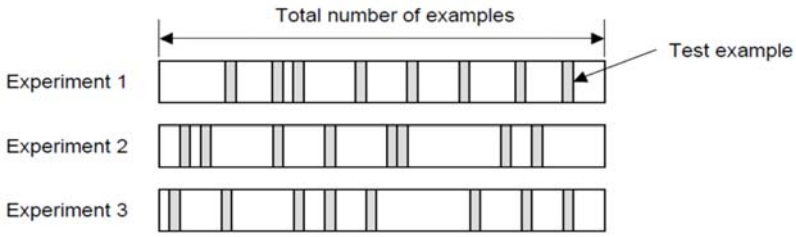

- Split dataset into two groups
  - Training set: used to train the classifier
  - Test set: used to estimate the error rate of the trained classifier
  - A typical application the holdout method is determining a stopping point for the back propagation error
- Drawbacks
  - In problems where we have a sparse dataset we may not be able to afford the "luxury" of setting aside a portion of the dataset for testing
  - Since it is a single train-and-test experiment, the holdout estimate of error rate will be misleading if we happen to get an "unfortunate" split
- The limitations of the hold-out can be overcome with a family of resampling methods at the expense of more computations






# Random Subsampling

- Random Subsampling performs K data splits of the dataset
  - Each split randomly selects a (fixed) no. examples without replacement
  - For each data split we retrain the classifier from scratch with the training examples and estimate $E_i$ with the test examples

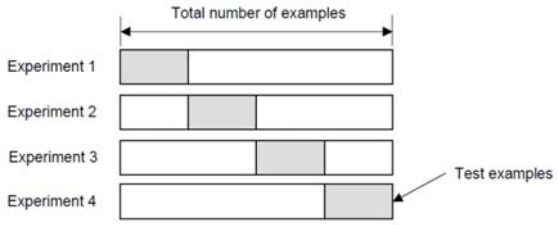



$$E = \frac{1}{K}\sum_{i=1}^{K} E_i$$

  - The true error estimate is obtained as the average of the separate estimates $E_i$
    - *This estimate is significantly better than the holdout estimate*

# K-Fold Cross-validation

- Create a K-fold partition of the dataset
  - For each of K experiments, use K-1 folds for training and the remaining one for testing




  - K-Fold Cross validation is similar to Random Subsampling
    - *The advantage of K-Fold Cross validation is that all the examples in the dataset are eventually used for both training and testing*
  - As before, the true error is estimated as the average error rate

.

$$E = \frac{1}{K}\sum_{i=1}^{K} E_i$$

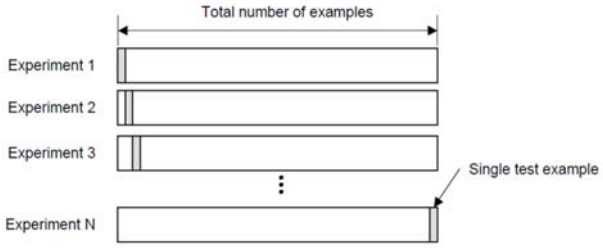


# How many folds are needed?

- With a large number of folds
  - The bias of the true error rate estimator will be small (very accurate)
  - The variance of the true error rate estimator will be large
  - The computational time will be very large as well (many experiments)
- With a small number of folds
  - The computation time are reduced because of a few of experiments
  - The variance of the estimator will be small
  - The bias of the estimator will be large (higher than the true error rate)
- In practice, the choice of the number of folds depends on the size of the dataset
  - For large datasets, even 3-Fold Cross Validation will be quite accurate
  - For very sparse datasets, we may have to use leave-one-out in order to train on as many examples as possible
- A common choice for K-Fold Cross Validation is K=10

# Leave-one-out Cross Validation

- Leave-one-out is the degenerate case of K-Fold Cross Validation, where K is chosen as the total number of examples
  - For a dataset with N examples, perform N experiments
  - For each experiment use N-1 examples for training and the remaining example for testing




- As usual, the true error is estimated as the average error rate on test examples

$$E = \frac{1}{N}\sum_{i=1}^{N} E_i$$

- Suit for small size of sample



# Bootstrap

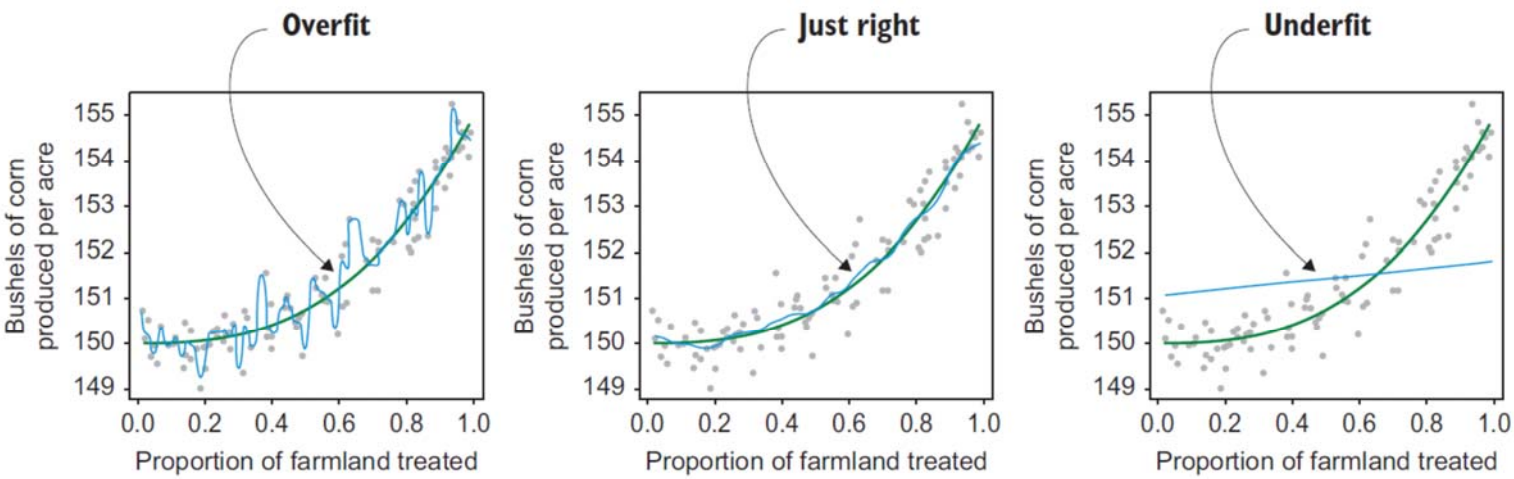
- Bootstrap techniques are relatively new to the field of statistics.
- The first use was published in a 1979 paper by Bradley Efron.
- As computing power has increased and becomes less expensive, bootstrap techniques have become more widespread.
- It suits for small size of dataset.
- Repeat random sample with the replacement.
  - Some data points may be duplicated, and others data points from the initial data may be omitted in a bootstrap sample.
  - With the help of a computer, thousands of bootstrap samples can be constructed in a relatively short time.

# Bootstrap example

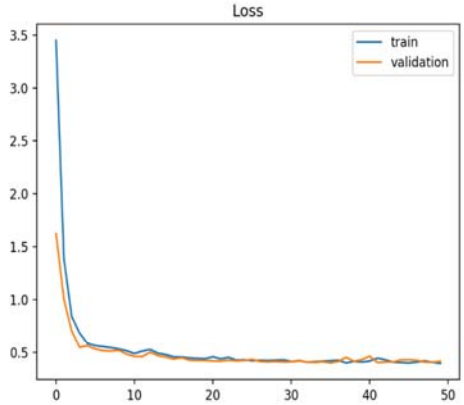
- If we have numerical example, [2, 4, 5, 6, 6, 7].
- We need 5 numbers. The bootstrap samples 10 times may be:
  - 2, 4, 5, 6, 7
  - 2, 2, 5, 5, 7
  - 4, 4, 5, 6, 7
  - 2, 5, 6, 7, 4
  - 4, 2, 5, 6, 7
  - 2, 4, 4, 5, 5
  - 5, 5, 5, 7, 7
  - 2, 4, 4, 5, 7
  - 4, 5, 5, 6, 6
  - 2, 4, 6, 6, 7

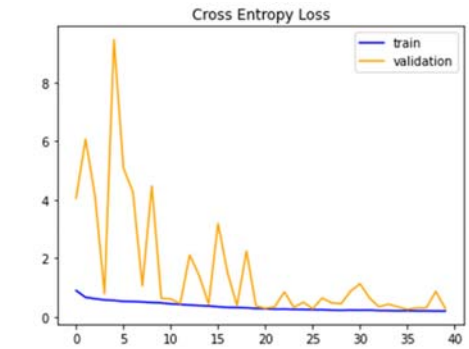


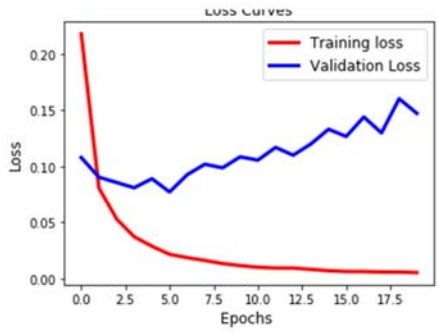
# Model Evaluation and Optimization

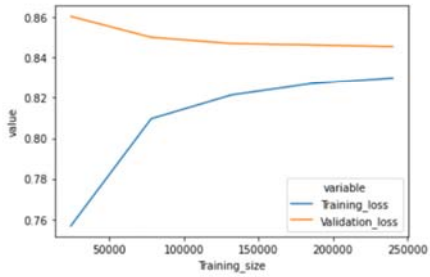


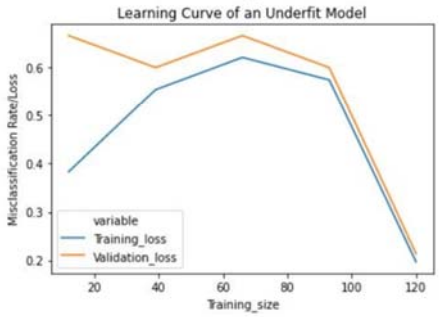
# Model Evaluation and Optimization



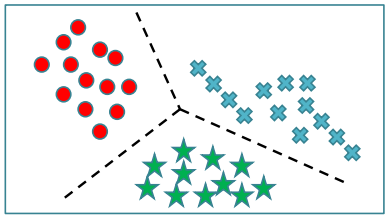





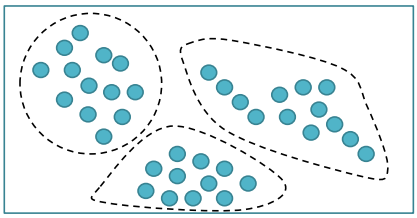

# Comparing Methods of Supervised and Unsupervised Learning

- Classification and clustering look similar
  - But the technology is quite different
- Classification and regression look quite different
  - But the technology is fairly similar

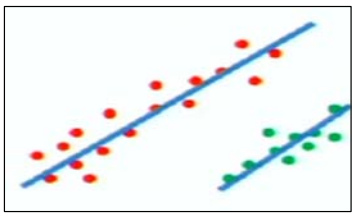
Classification



Clustering



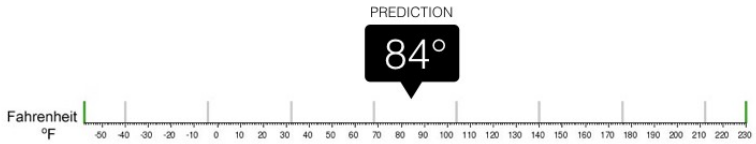
Regression



# Classification VS Regression

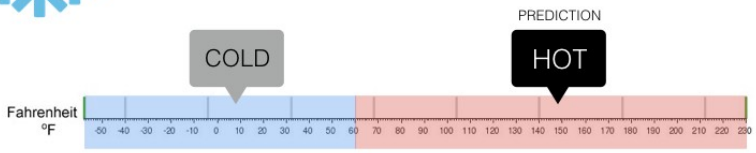
**Regression**

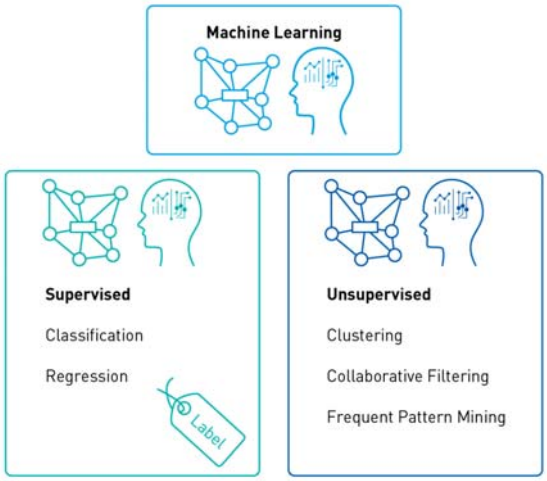
What is the temperature going to be tomorrow?



**Classification**

Will it be Cold or Hot tomorrow?






# Classification VS Regression

Team → Classification → Predicting **Win or Lose** Yes or No

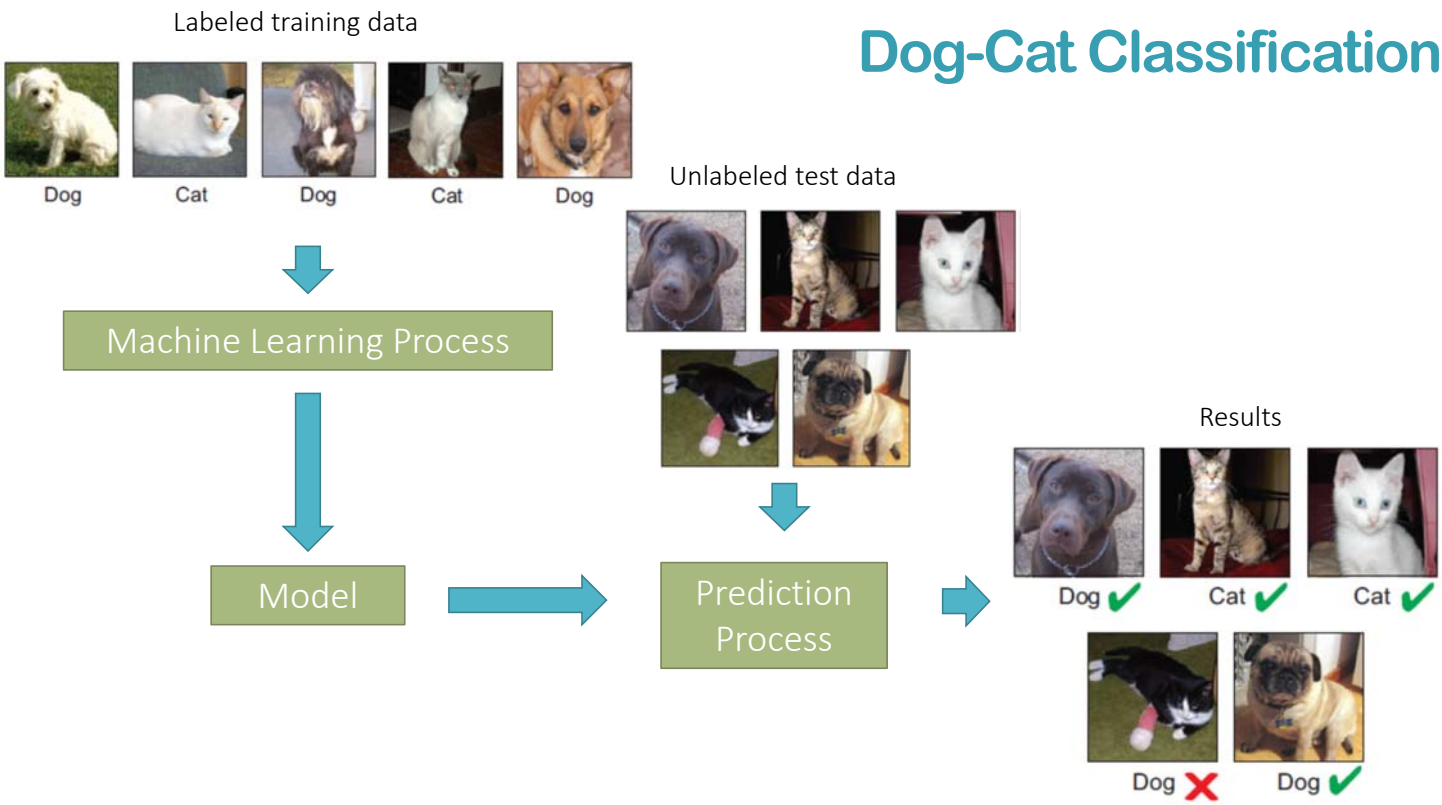
Team → Regression → Forecasting **Chance of winning** Probability or Percentage

# Classification VS Regression

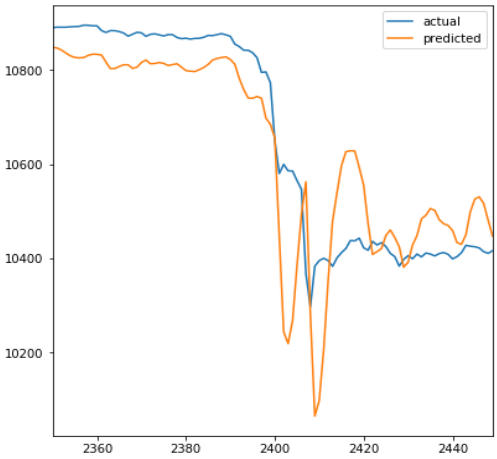
- Classification is different from regression:
  - Classification : predicting a discrete class label
  - Regression : predicting a continuous quantity
- Some overlap between the algorithms for classification and regression:
  - A classification may predict a continuous value but in the form of a probability for a class label
  - A regression may predict a discrete value but in the form of an integer quantity
- Some algorithms can be used for both classification and regression with small modifications:
  - decision trees and artificial neural networks
- Some algorithms cannot, or cannot easily be used for both types:
  - Linear regression for regression
  - Logistic regression for classification
- Importantly, evaluate classification and regression varies and do not overlap:
  - Classification: evaluate using accuracy
  - Regression: evaluated using root mean squared error or mean absolute error

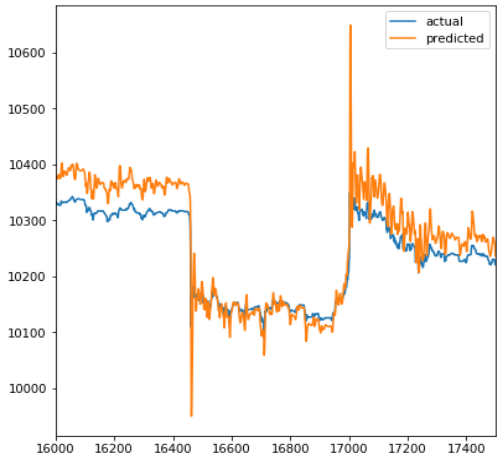


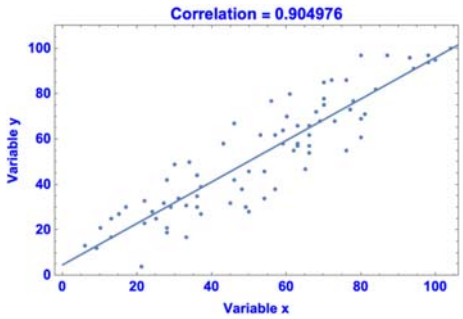
# Dog-Cat Classification



# Regression Example

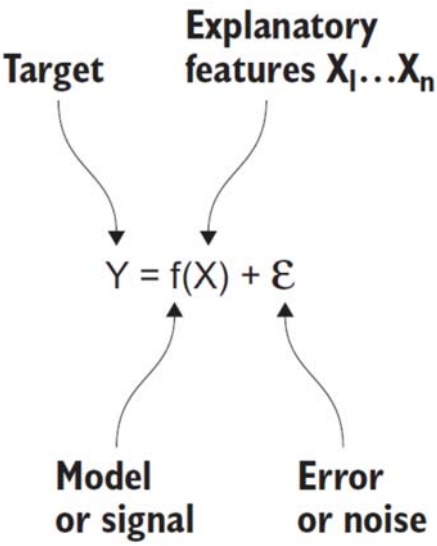





# Model Generation and Prediction




Model Generation

$$Y_{pred} = f_{est}(\mathbf{X}_{new})$$

Prediction

# Classification

- Classification predictive modelling is the task of approximating a mapping function ($f$) from input variables ($X$) to discrete output variables ($y$).

$$y = f(X)$$

- The output variables are often called labels or categories. The mapping function predicts the class or category for a given observation.
- For example, an email of text can be classified as belonging to one of two classes: "spam" and "not spam".
  - A classification problem requires that examples be classified into one of two or more classes.
  - A classification can have real-valued or discrete input variables.
  - A problem with two classes is often called a two-class or binary classification problem.
  - A problem with more than two classes is often called a multi-class classification problem.
  - A problem where an example is assigned multiple classes is called a multi-label classification problem.



# Classification

- Classification is a process of categorizing a given set of data into classes
- The classes are often referred to as target, label or categories
  - $C = (C1, C2 \ldots . Cm)$
  - *(Iris Setosa & Iris versicolor in iris flower dataset)*
- Each data sample is represented by an n-dimensional feature vector,
  - $X = (x1, x2, \ldots\ xn)$
  - *(sepal_width, sepal_length, petal_width, petal_length in iris flower dataset)*
- The classifier calculates the probability of data sample (unknown/new) w.r.t each class/category.
- The classifier decides the class on the basis of highest priority

# Classification

- For example, Heart disease detection can be identified as a classification problem,
  - This is a binary classification since there can be only two classes i.e.
    - *Class1: has heart disease*
    - *Class2: does not have heart disease.*
- The classifier, in this case, needs training data to understand how the given input variables are related to the class.
- Once the classifier is trained accurately, it can be used to detect whether heart disease is there or not for a particular patient.



# Regression

- Regression predictive modelling is the task of approximating a mapping function (f) from input variables (X) to a continuous output variable (y).

$$y = f(X)$$

- A continuous output variable is a real-value, such as an integer or floating point value. These are often quantities, such as amounts and sizes.
- For example, a house may be predicted to sell for a specific dollar value, perhaps in the range of $100,000 to $200,000.
  - A regression problem requires the prediction of a quantity.
  - A regression can have real valued or discrete input variables.
  - A problem with multiple input variables is often called a multivariate regression problem.
  - A regression problem where input variables are ordered by time is called a time series forecasting problem.

# Evaluation in CS

- **Design of algorithms**
  - Comparing the efficiency of different algorithms for the same problem
  - Time and space measures
  - Big-O notation for expressing time complexity
- AI/Data Science
  - Time measures/ time complexity
  - Confusion matrix
    - *Accuracy, Precision, Recall, F-measure (F1-score)*
    - *AUC, ROC curve*
    - *mAP*
  - Forecasting
    - *RMSE, MAPE*
  - Confidence



# Confusion Matrix for two categories

| Predict (Conclusion/ Decision) | *Actual Value* (Population Condition /Truth) | | | |
|---|---|---|---|---|
| | *Positive* | *Negative* | | |
| Positive | True Positive (TP) (Correct decision) Hit | False Positive (FP) (Type I error) false alarm, overestimation | $\sum Predicted\ Positive$ | $Precision = \frac{TP}{\sum Predicted\ Positive}$ |
| Negative | False Negative, FN (Type II error) miss, underestimation | True Negative (TN) (Correct rejection) | $\sum Predicted\ Nagative$ | |
| | $\sum Condition\ Positive$ | $\sum Condition\ Nagative$ | $\sum All$ | |
| | Recall, Sensitivity, Hit rate, TP Rate $= \frac{TP}{\sum Actual\ Positive}$ | Specificity, TN rate $= \frac{TN}{\sum Actual\ Negative}$ | | Accuracy $= \frac{TP + TN}{\sum All}$ |
| | Miss rate, FN Rate $= \frac{FN}{\sum Actual\ Positive}$ | | | |

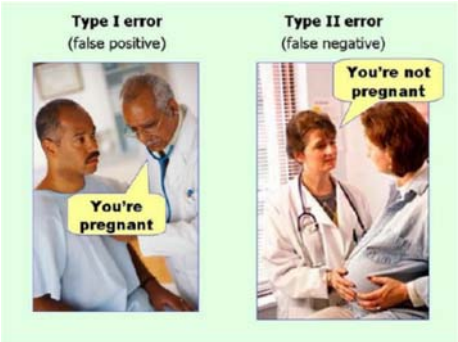
# Example Evaluation

| Index | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Actual | Dog | Dog | Dog | Not Dog | Dog | Not Dog | Dog | Dog | Not Dog | Not Dog |
| Predicted | Dog | Not Dog | Dog | Not Dog | Dog | Dog | Dog | Dog | Not Dog | Not Dog |
| Result | TP | FN | TP | TN | TP | FP | TP | TP | TN | TN |

- Actual Dog Counts = 6
- Actual Not Dog Counts = 4
- True Positive Counts = 5
- False Positive Counts = 1
- True Negative Counts = 3
- False Negative Counts = 1

| | | Actual | | |
|---|---|---|---|---|
| | | Dog | Not Dog | |
| Predicted | Dog | True Positive (TP=5) | False Positive (FP=1) | Precision (PR=5/6) |
| | Not Dog | False Negative (FN=1) | True Negative (TN=3) | |
| | | Recall (RC=5/6) | | Accuracy (AC=8/10) |



# Examples of Type I & Type II errors

- U.K. police defend shoot-to-kill after mistake (MSNBC, July 24th).
  - Blair said Menezes had emerged from an apartment block in south London that had been under surveillance in connection with Thursday's attacks and refused police orders to halt. Menezes had also been wearing an unseasonably heavy coat, further raising police suspicions.
- Email Filtering
  - A message is blocked although it is not spam.
  - A message is unblocked although it is spam.